

Fewer flood deaths. More flood wanters.

Flood deaths per capita in the US have fallen more than 60% since the 1940s. Over the same stretch, Americans have been moving into floodplains, not away from them. Both are true — and neither alone tells you whether we're actually safer.

–65%

decline in per-capita weather deaths, 1940s to 2010s

86 yrs

of NOAA fatality records analyzed, 1940–2025

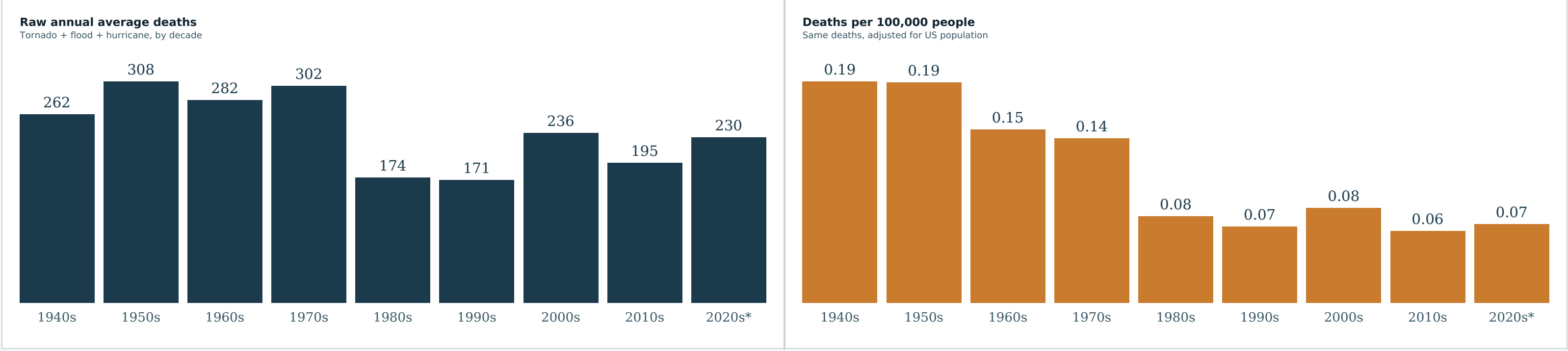
1 metric

problem: "death rate" isn't one number

01 — THE RAW NUMBERS

Deaths are down. Population is way up.

Using NOAA's 81-Year Hazard Statistics, combined annual average fatalities from tornadoes, floods, and hurricanes hover in the low 200s across most decades, with no clear long-run upward trend. But that number alone hides the more important story.

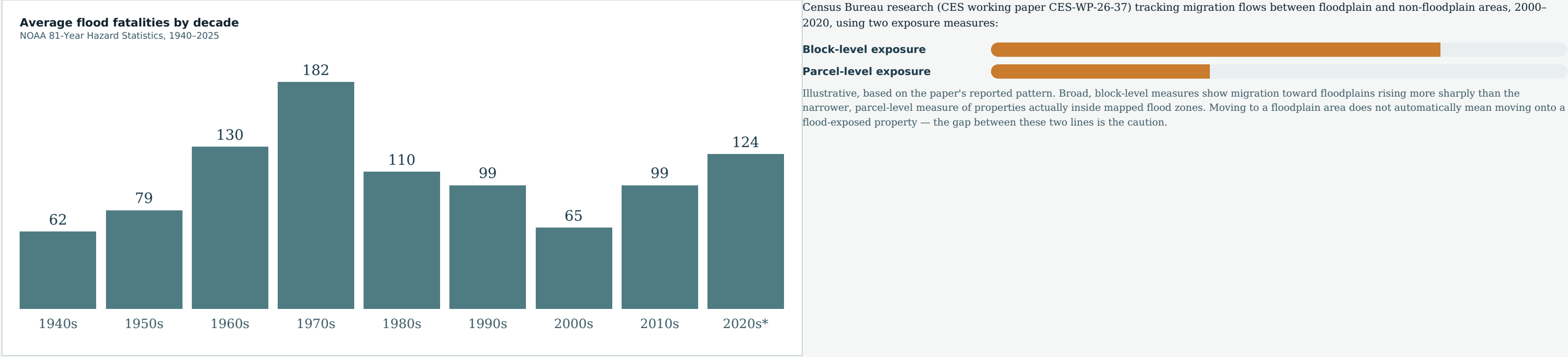


The denominator matters. Raw deaths in the 2000s (236/yr, driven by Hurricane Katrina) look similar to the 1940s (262/yr). But the US population nearly tripled in that span — so the per-capita risk of dying from these hazards fell by roughly two-thirds. Adaptation — building codes, warning systems, evacuation infrastructure — is doing real work, even where the raw death toll looks flat.

02 — WHERE PEOPLE ARE MOVING

Flooding: the hazard I chose

Flood fatalities specifically show more volatility than tornado or hurricane deaths, spiking in years with major inland flooding (1972's Hurricane Agnes, 1993's Midwest floods, 2024's Hurricane Helene) rather than trending smoothly. So a natural next question: are Americans moving toward flood risk, or away from it?



03 — A MEASUREMENT CAUTION

Not every "death count" measures the same thing

A recent NBER paper (Burke et al., 2025) estimates ambient temperature is responsible for 5–12% of all deaths across a 30-country sample — hundreds of thousands of deaths a year in the US and EU alone. That dwarfs any NOAA hazard category, including flooding.

These numbers aren't in tension with the flood chart above — they aren't measuring the same thing. NOAA's flood fatalities are deaths directly and immediately caused by a named event. Burke's temperature-attributable mortality is a statistical estimate of excess deaths, largely cardiovascular and respiratory, linked to temperature deviation across an entire population over a full year, most of which never get coded as a "weather death" at all. Before comparing any two mortality statistics in this space, ask what each one is actually counting.

04 — THE ECONOMICS

Adaptation isn't free — someone pays, and someone decides

Levees, elevated foundations, flood insurance mandates, and relocating aging infrastructure to higher ground all cost money, a real tradeoff against other public spending. Nordhaus-style estimates of coastal capital relocation suggest the annual cost of gradually moving vulnerable structures to safer ground is a small share of GDP in aggregate, but far from small to the households and municipalities who bear it directly.

Floodplain migration raises the question from the other direction: if moving toward water is individually rational because protection has improved, who bears the residual risk when protection fails? Subsidized flood insurance and disaster relief effectively socialize part of that risk — which changes the incentive to move there in the first place. That's a classic externality problem, not just a safety story.

05 — OH, WAIT

What this data can't tell you

A falling per-capita death rate is easy to read as "we've solved flood risk." Hold on: falling deaths could reflect real adaptation, or it could reflect that the worst-exposed structures simply haven't been tested yet by a large enough event. The 2020s figure above (123.5 flood deaths/year average) is already trending back toward 1960s–70s levels after decades of decline, built on just six years of data. One or two catastrophic seasons could reverse the story this poster tells. Falling death rates measure what has happened, not what protection actually exists against what hasn't happened yet.

Sources

- NOAA National Weather Service, 81-Year Hazard Statistics (1940–2025) — weather.gov/media/hazstat/81year_2025.pdf
- U.S. Census Bureau, Historical national population estimates
- U.S. Census Bureau CES Working Paper CES-WP-26-37, Migration flows and floodplain exposure, 2000–2020
- Burke, M. et al. (2025), "Understanding and Addressing Temperature Impacts on Mortality," NBER Working Paper 34313

